



Wine Quality Prediction

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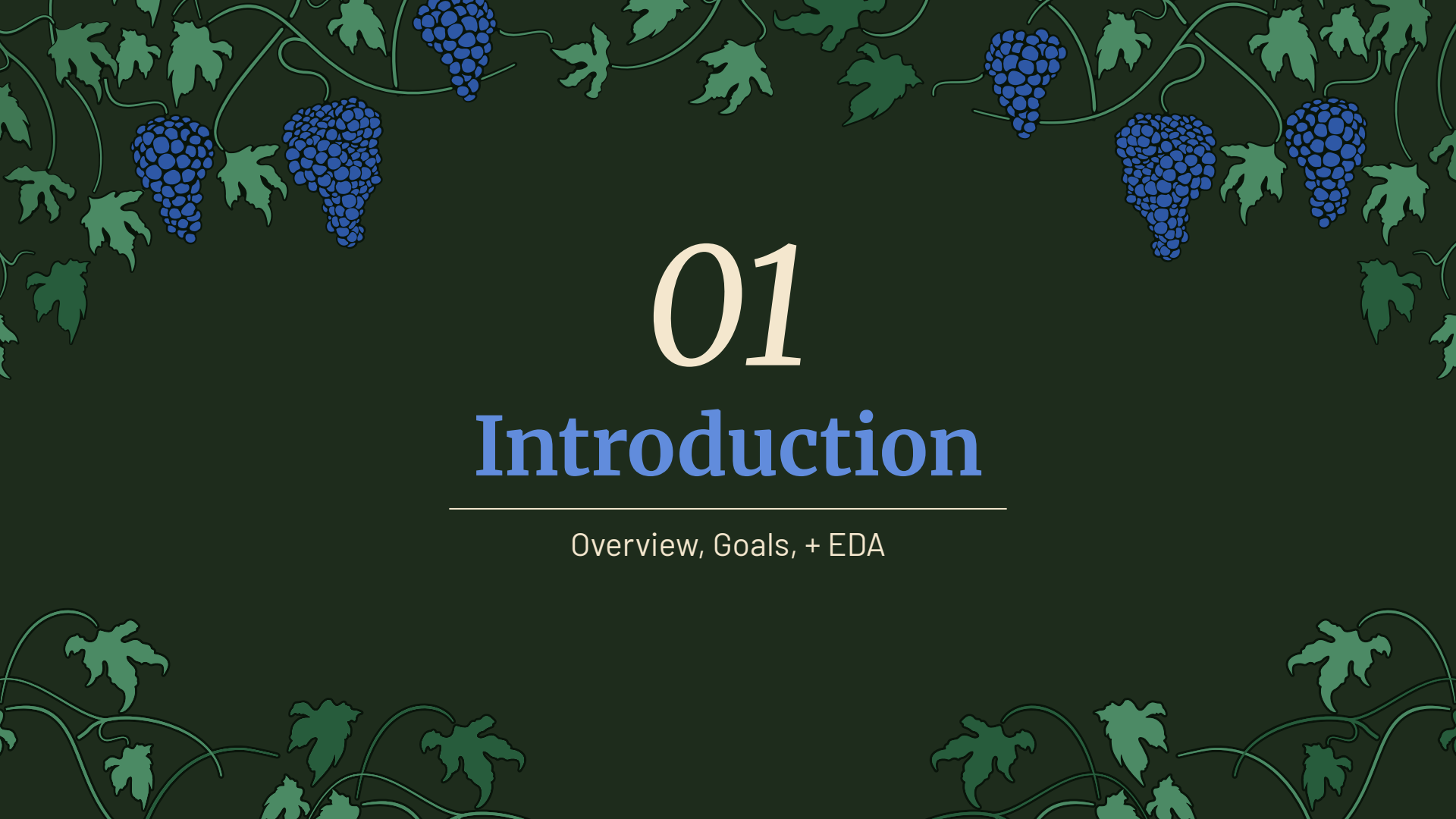
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Conclusion

Which is the best
model?



01

Introduction

Overview, Goals, + EDA

Overview



- Wine Quality data from UCI Machine Learning Repository
- Analyzed physicochemical properties of wine to predict wine quality scores
- 2 datasets:
 - White wine (4,898 observations)
 - Red wine (1,599 observations)
- Both have 11 predictor variables and 1 response variable (quality)



Our Goal

Predict wine quality scores using 3 different methods (linear regression, regression trees, and random forests) to find which is best at predicting wine quality, lastly, compare which model has the best overall predictive accuracy.

Summary Statistics

```
> summary(white)
```

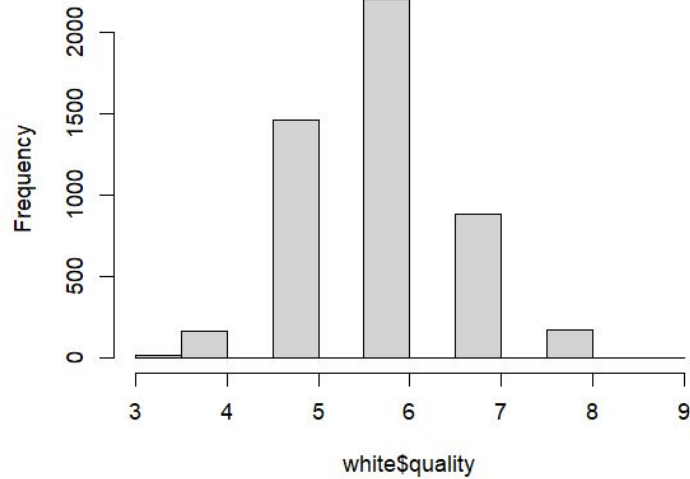
fixed.acidity	volatile.acidity	citric.acid
Min. : 3.800	Min. :0.0800	Min. :0.0000
1st Qu.: 6.300	1st Qu.:0.2100	1st Qu.:0.2700
Median : 6.800	Median :0.2600	Median :0.3200
Mean : 6.855	Mean :0.2782	Mean :0.3342
3rd Qu.: 7.300	3rd Qu.:0.3200	3rd Qu.:0.3900
Max. :14.200	Max. :1.1000	Max. :1.6600
residual.sugar	chlorides	free.sulfur.dioxide
Min. : 0.600	Min. :0.00900	Min. : 2.00
1st Qu.: 1.700	1st Qu.:0.03600	1st Qu.: 23.00
Median : 5.200	Median :0.04300	Median : 34.00
Mean : 6.391	Mean :0.04577	Mean : 35.31
3rd Qu.: 9.900	3rd Qu.:0.05000	3rd Qu.: 46.00
Max. :65.800	Max. :0.34600	Max. :289.00
total.sulfur.dioxide	density	pH
Min. : 9.0	Min. :0.9871	Min. :2.720
1st Qu.:108.0	1st Qu.:0.9917	1st Qu.:3.090
Median :134.0	Median :0.9937	Median :3.180
Mean :138.4	Mean :0.9940	Mean :3.188
3rd Qu.:167.0	3rd Qu.:0.9961	3rd Qu.:3.280
Max. :440.0	Max. :1.0390	Max. :3.820
sulphates	alcohol	quality
Min. :0.2200	Min. : 8.00	Min. :3.000
1st Qu.:0.4100	1st Qu.: 9.50	1st Qu.:5.000
Median :0.4700	Median :10.40	Median :6.000
Mean :0.4898	Mean :10.51	Mean :5.878
3rd Qu.:0.5500	3rd Qu.:11.40	3rd Qu.:6.000
Max. :1.0800	Max. :14.20	Max. :9.000

```
> summary(red)
```

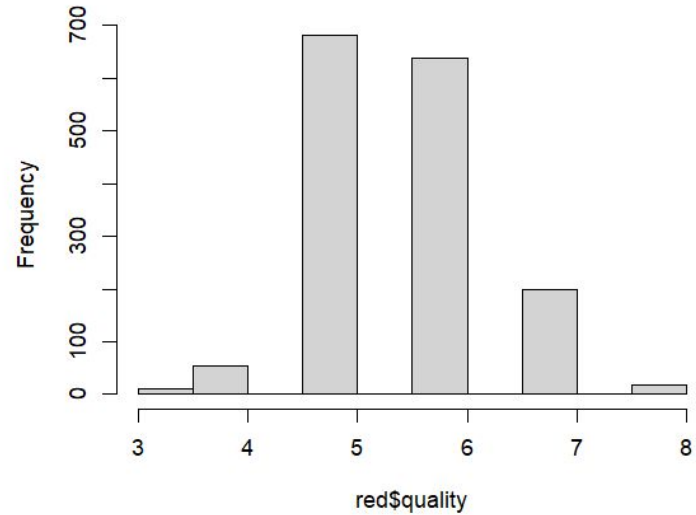
fixed.acidity	volatile.acidity	citric.acid
Min. : 4.60	Min. :0.1200	Min. :0.000
1st Qu.: 7.10	1st Qu.:0.3900	1st Qu.:0.090
Median : 7.90	Median :0.5200	Median :0.260
Mean : 8.32	Mean :0.5278	Mean :0.271
3rd Qu.: 9.20	3rd Qu.:0.6400	3rd Qu.:0.420
Max. :15.90	Max. :1.5800	Max. :1.000
residual.sugar	chlorides	free.sulfur.dioxide
Min. : 0.900	Min. :0.01200	Min. : 1.00
1st Qu.: 1.900	1st Qu.:0.07000	1st Qu.: 7.00
Median : 2.200	Median :0.07900	Median :14.00
Mean : 2.539	Mean :0.08747	Mean :15.87
3rd Qu.: 2.600	3rd Qu.:0.09000	3rd Qu.:21.00
Max. :15.500	Max. :0.61100	Max. :72.00
total.sulfur.dioxide	density	pH
Min. : 6.00	Min. :0.9901	Min. :2.740
1st Qu.: 22.00	1st Qu.:0.9956	1st Qu.:3.210
Median : 38.00	Median :0.9968	Median :3.310
Mean : 46.47	Mean :0.9967	Mean :3.311
3rd Qu.: 62.00	3rd Qu.:0.9978	3rd Qu.:3.400
Max. :289.00	Max. :1.0037	Max. :4.010
sulphates	alcohol	quality
Min. :0.3300	Min. : 8.40	Min. :3.000
1st Qu.:0.5500	1st Qu.: 9.50	1st Qu.:5.000
Median :0.6200	Median :10.20	Median :6.000
Mean :0.6581	Mean :10.42	Mean :5.636
3rd Qu.:0.7300	3rd Qu.:11.10	3rd Qu.:6.000
Max. :2.0000	Max. :14.90	Max. :8.000

Histogram of Quality

Histogram of white\$quality



Histogram of red\$quality





02

Method 1: Linear Regression

Benchmark Model

```
> summary(model_white)
```

Call:

```
lm(formula = quality ~ ., data = white_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-3.6485	-0.4948	-0.0338	0.4902	3.0091

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.674e+02	4.194e+01	6.376	2.45e-10	***
fixed.acidity	1.833e-01	4.298e-02	4.266	2.13e-05	***
volatile.acidity	-1.745e+00	2.139e-01	-8.159	7.34e-16	***
citric.acid	-5.153e-02	1.656e-01	-0.311	0.755741	
residual.sugar	1.076e-01	1.573e-02	6.842	1.15e-11	***
chlorides	-1.116e+00	9.594e-01	-1.163	0.244946	
free.sulfur.dioxide	6.703e-03	1.768e-03	3.790	0.000157	***
total.sulfur.dioxide	4.963e-05	7.069e-04	0.070	0.944039	
density	-2.708e+02	4.250e+01	-6.373	2.50e-10	***
pH	1.535e+00	2.100e-01	7.309	4.48e-13	***
sulphates	8.838e-01	1.995e-01	4.429	1.02e-05	***
alcohol	7.574e-02	5.165e-02	1.467	0.142732	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.7812 on 1427 degrees of freedom
Multiple R-squared: 0.3038, Adjusted R-squared: 0.2985
F-statistic: 56.61 on 11 and 1427 DF, p-value: < 2.2e-16

```
> summary(model_red)
```

Call:

```
lm(formula = quality ~ ., data = red_train)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.64994	-0.37110	-0.05955	0.45154	2.00330

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.452e+01	2.250e+01	1.090	0.276	
fixed.acidity	2.087e-02	2.789e-02	0.748	0.455	
volatile.acidity	-1.084e+00	1.286e-01	-8.425	< 2e-16	***
citric.acid	-1.916e-01	1.572e-01	-1.219	0.223	
residual.sugar	1.501e-02	1.637e-02	0.917	0.359	
chlorides	-1.847e+00	4.447e-01	-4.154	3.46e-05	***
free.sulfur.dioxide	3.302e-03	2.268e-03	1.456	0.146	
total.sulfur.dioxide	-3.058e-03	7.646e-04	-3.999	6.69e-05	***
density	-2.032e+01	2.298e+01	-0.885	0.377	
pH	-4.211e-01	2.039e-01	-2.066	0.039	*
sulphates	8.651e-01	1.203e-01	7.189	1.05e-12	***
alcohol	2.740e-01	2.838e-02	9.654	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6508 on 1427 degrees of freedom
Multiple R-squared: 0.3529, Adjusted R-squared: 0.348
F-statistic: 70.76 on 11 and 1427 DF, p-value: < 2.2e-16

Best Subsets Variable Selection

White Wine

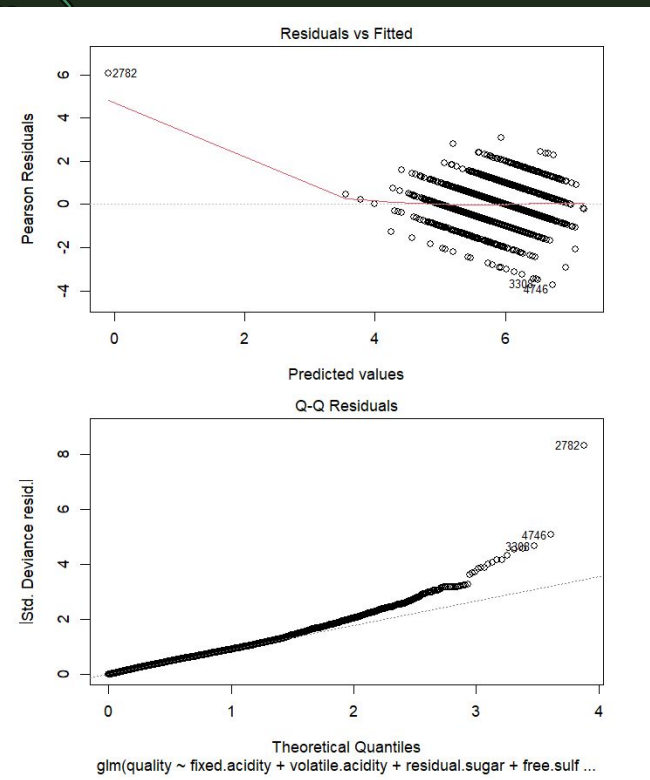
- Used BIC criterion → 7 predictors
 - Fixed acidity, volatile acidity, residual sugar, free sulfur dioxide, density, pH, and sulfates
- In-sample MSE = 0.6101
- 5-fold Cross Validation = 0.5741
 - Aka out of sample MSE
- Adjusted R-squared = 0.2987

Red Wine

- Used BIC criterion → 6 predictors
 - Volatile acidity, chlorides, total sulfur dioxide, pH, sulfates, and alcohol
- In-sample MSE = 0.4239
- 5-fold Cross Validation = 0.4243
 - Aka out of sample MSE
- Adjusted R-squared = 0.3474

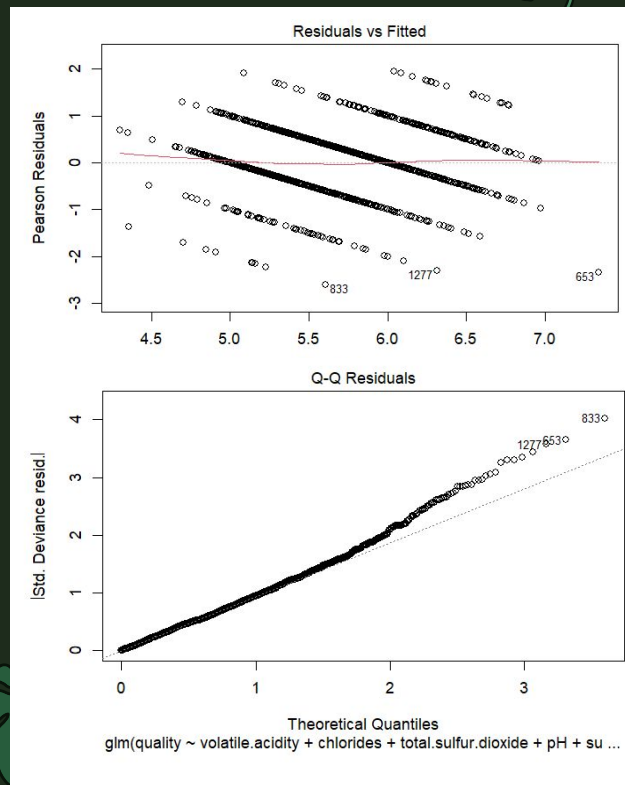
Cross Validation/Residual Plots

White Wine



- Not randomly distributed around 0 (shows pattern)
- Q-Q plot residuals do NOT fall on dotted line
 - Tail on white wine shows extreme outliers present
- Linear Regression model likely not a good fit
 - Normality assumption violated

Red Wine

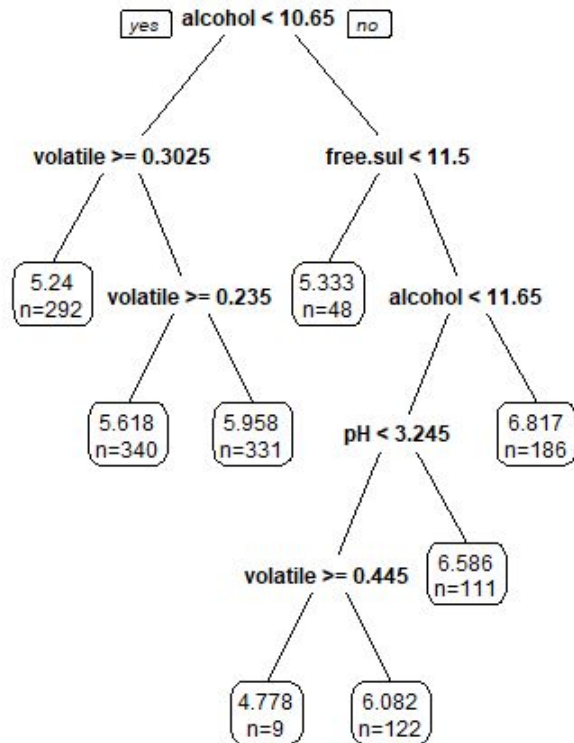




03

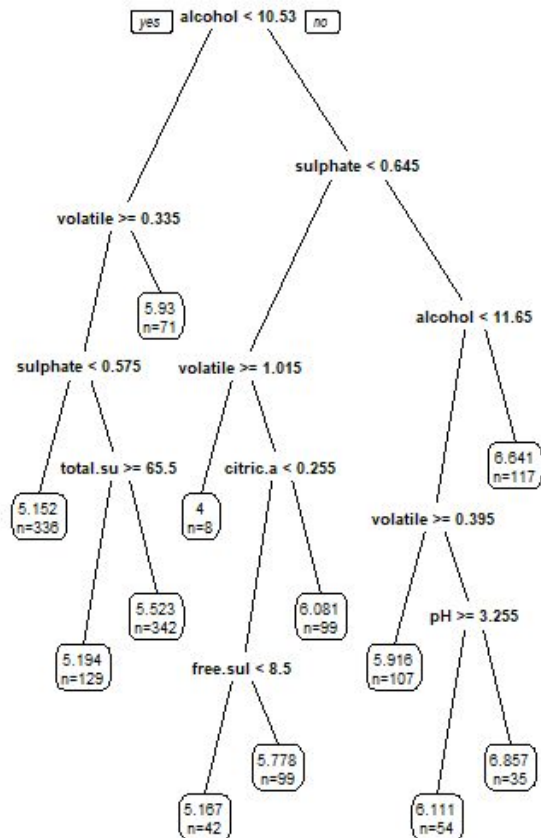
Method 2: Regression Tree

White Wine Tree



- MSE (in sample) = 0.5956
- MSPE (out of sample) = 0.5743
- Highest predicted quality (6.817)
 - Wines w/ high alcohol, moderate free sulfur, and alcohol < 11.65
- Lowest predicted quality (4.778)
 - Wines w/ lower alcohol and high volatile acidity > 0.445

Red Wine Tree



- MSE (in sample) = 0.4001
- MSPE (out of sample) = 0.4175
- Highest predicted quality (6.857)
 - Wines w/ high alcohol, low sulphates, high volatile acidity + pH > 3.255
- Lowest predicted quality (4)
 - Wines w/ high alcohol, low sulfates, + high volatile acidity > 1.015



04

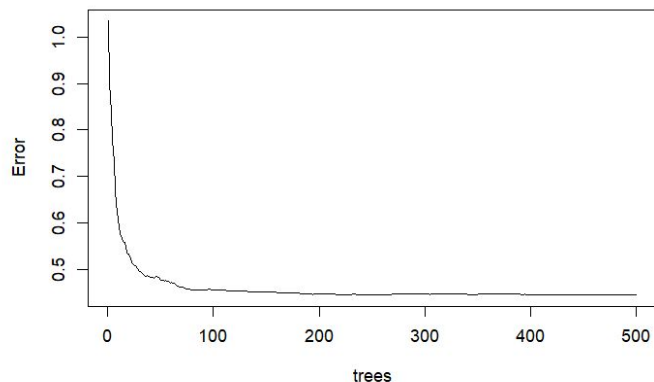
Method 3: Random Forests

Random Forest Model

```
Call:
randomForest(formula = quality ~ ., data = white_train, ntree = 500)
  Type of random forest: regression
    Number of trees: 500
No. of variables tried at each split: 3
```

Mean of squared residuals: 0.4439656
% Var explained: 48.93

rf_white

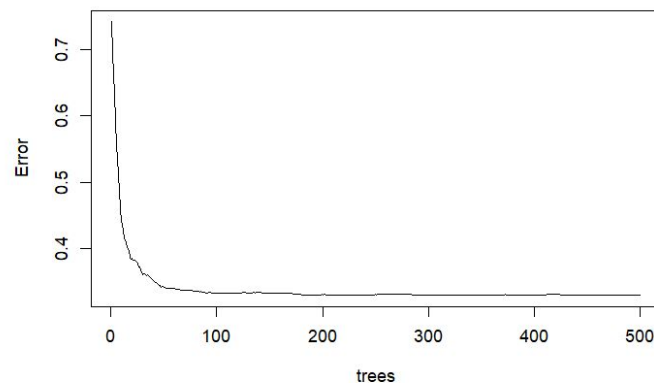


Out of sample MSE = **0.5337**

```
Call:
randomForest(formula = quality ~ ., data = red_train, ntree
= 500)
  Type of random forest: regression
    Number of trees: 500
No. of variables tried at each split: 3
```

Mean of squared residuals: 0.3307936
% Var explained: 49.04

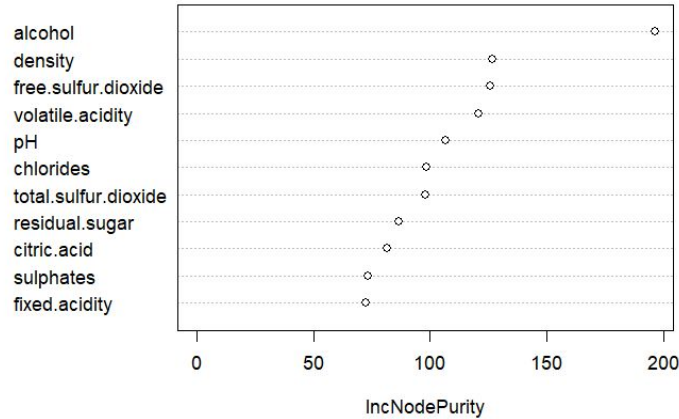
rf_red



Out of sample MSE = **0.2774**

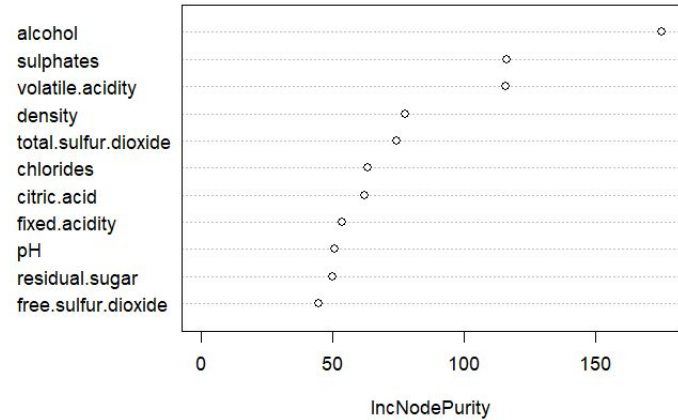
Variable Importance Plots

rf_white



- Alcohol most important
- Density and free sulfur dioxide have moderate importance
- Remaining less significant

rf_red

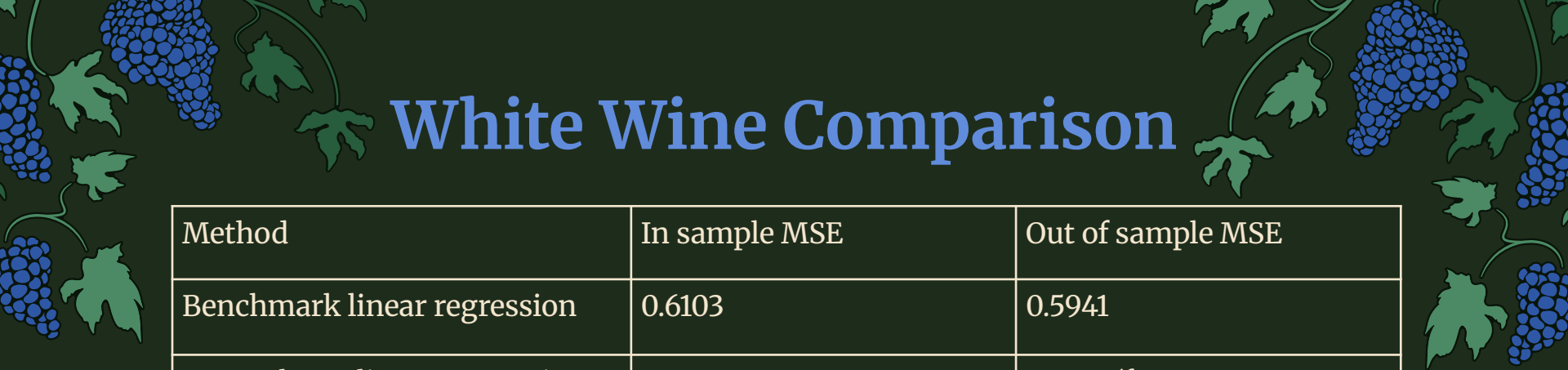


- Alcohol most important
- Sulphates and volatile acidity have moderate importance
- Remaining less significant



05

Results Overview



White Wine Comparison

Method	In sample MSE	Out of sample MSE
Benchmark linear regression	0.6103	0.5941
Best subsets linear regression	0.6101	0.5741 (from cross validation)
Regression Tree	0.5956	0.5743
Random Forest	0.4439	0.5337

- Random Forest method produces lowest in and out of sample MSEs



Red Wine Comparison

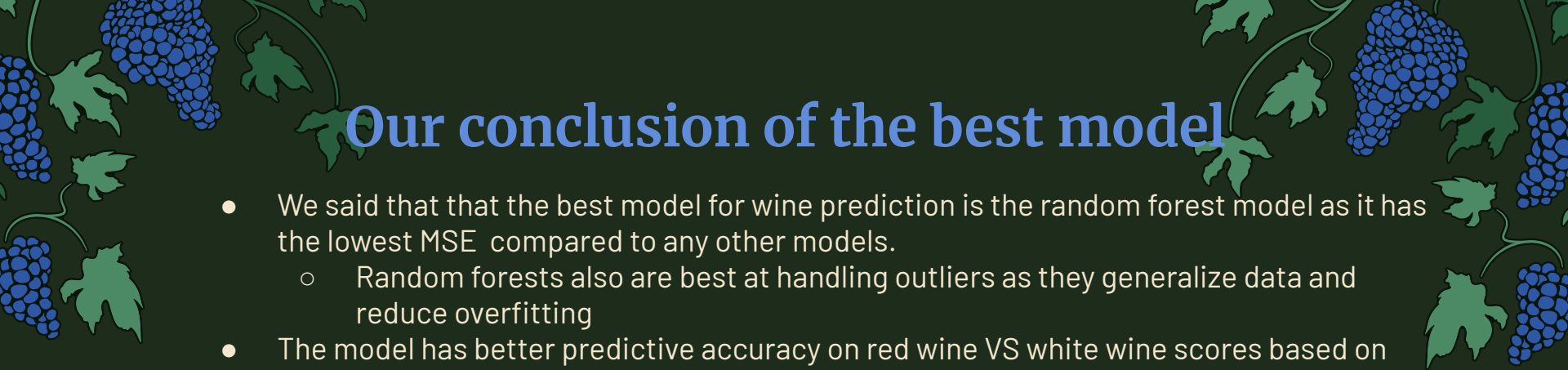
Method	In sample MSE	Out of sample MSE
Benchmark linear regression	0.4236	0.3906
Best subsets linear regression	0.4239	0.4243 (from cross validation)
Regression Tree	0.4006	0.4175
Random Forest	0.3308	0.2774

- Random Forest method produces lowest in and out of sample MSEs



05

Conclusion



Our conclusion of the best model

- We said that that the best model for wine prediction is the random forest model as it has the lowest MSE compared to any other models.
 - Random forests also are best at handling outliers as they generalize data and reduce overfitting
- The model has better predictive accuracy on red wine VS white wine scores based on lower MSE scores.

Random Forest MSE results

	White wine	Red wine
MSE (in sample)	0.4439	0.3308
MSE (out of sample)	0.5337	0.2774



Thank You!